

Adam Rolander

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EXPERIENCE

AI Engineer

UCSD Extended Reality Lab | La Jolla, CA | January 2025 - Present

- ❖ Implementing a **3D asset diffusion transformer** in a custom pipeline, **increasing virtual environment creation productivity by 10x**
- ❖ Developing a **scenic 3D worldbuilding pipeline** from a single image with **segmentation, object classification, and 3D asset placement** via a custom monocular depth estimation model
- ❖ Leading **AI-assisted 3D environment generation**, including automated MetaHuman creation and diffusion-transformer-based landscape maps from textual descriptions
- ❖ Training **Hidden Markov Models** to quantify path similarity in multi-subject experimental movement data

Data Science Intern

Cornerstone Whole Healthcare Organization | Coeur d'Alene, ID (Remote) | June - September 2025

- ❖ Built and deployed an **LSTM forecasting model for clinical costs** with **TensorFlow.js**, supporting superior operational insights and value-based care adoption
- ❖ Analyzed the **OHA 988 Crisis Line dataset** (~200k rows x ~250 columns) with **data cleaning, EDA, visualization, and NLP preprocessing** of caller presenting issues
- ❖ Developed **Random Forest and XGBoost models** for suicidality risk and call duration, applied **SMOTE**, and presented findings; insights are now **used statewide** to optimize call center operations and improve outcomes

Software Engineering Intern

Strix Leviathan | Seattle, WA (Remote) | June - September 2024

- ❖ Developed a **RESTful/Websocket client** for cryptocurrency market data, reducing API costs by **\$17,000/month**
- ❖ Automated **Docker deployments, integrated CI/CD pipelines** using **Pytest, Bash, AWS Code Artifact/ECR, and GitHub Actions**
- ❖ Researched **BTC market sentiment** and implemented **rolling-average-based trading strategies** using Python and Pandas for data analysis

EDUCATION

University of California, San Diego | La Jolla, CA

B.S. in Machine Learning & Mathematics | GPA - 4.0 | Expected Graduation June 2028

- ❖ **Relevant Coursework** - Deep Learning for NLP, ML Algorithms, PCA & Spectral Analysis, Data Structures
- ❖ **Awards & Activities** - Blackstone LaunchPad Incubator, National Merit Scholar, Triton Software Engineering

TECHNICAL SKILLS

Languages - Python, Java, C, Bash | **Machine Learning** - PyTorch, TensorFlow/Keras, FAISS, NumPy, Pandas, scikit-learn, Matplotlib | **Infrastructure** - AWS (ECR/ECS, CodeArtifact), Docker, Git/GitHub | **3D/XR Tools** - Unreal Engine, Blender, RealityCapture/iTwinCapture | **Data/Dev Tools** - MATLAB, Jupyter Notebook, Flutter, MERN Stack

PROJECTS

- 🌀 **Multimodal 3D Asset Retrieval** - Compared supervised and unsupervised methods for text-to-3D retrieval on the CAP3D dataset, pitting a sentence encoder baseline against a fine-tuned contrastive model (MiniLM-L6-V2 & ViT-B/16). Found that paraphrase-based data augmentation prevented supervised model overfitting, restored Stability, and halved Discrimination's false-positive rate
- 🌀 **Longitudinal Mental Health Analysis** - Implemented Random Forest Regression and Support Vector Machines on the Add Health dataset to study the impact of early-life emotional support on adult mental health; found **higher parental care predicts lower adult depression**
- 🌀 **Price Forecasting with LSTM Neural Networks** - Forecasted Binance price tick directions from high-frequency Limit Order Book data, achieving **66% accuracy and 1st place in competitive symposium**
- 🌀 **Brain Tumor Classification with Convolutional Neural Networks** - Performed multi-class brain tumor classification on augmented MRI data; achieved **96% accuracy** and published results as an **ISEF finalist**